

Recap.

- ① For continuous RV X with pdf $p_X(x)$ and RV $Y=g(X)$, $P_Y(y) = P_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$.
- ② For dataset $x_1, x_2, \dots, x_n$, assume they come from a distribution with parameter θ .
MLE $\hat{\theta}_{MLE} = \operatorname{argmax} L(\theta|x)$ finds the $\hat{\theta}$ making the function $L(\theta|x)$ achieves maximum value.
 $L(\theta|x) = P(x_1, x_2, \dots, x_n | \theta)$, "Likelihood of θ given x ".
e.g. Flipping a coin 10 times, observed 7 heads and 3 tails.
Define θ as probability of "head shows up", $L(\theta|x) = \theta^7(1-\theta)^3$, $\hat{\theta} = 0.7$.
- ③ Bias = $E(\hat{\theta}) - \theta$. $\hat{\theta}$ is unbiased estimator of θ .

Exercises.

Ex 1. Hint: For $X \sim \text{Exp}(\lambda)$, $p_X(x) = \lambda e^{-\lambda x}$ ($x \geq 0$), $E(X) = \frac{1}{\lambda}$, $\text{var}(X) = \frac{1}{\lambda^2}$.

$$\begin{aligned} \text{(i)} \quad X_i \sim \text{Exp}(\lambda), \quad \phi_{X_i}(t) &= E(e^{itX_i}) = \int_0^\infty e^{itx} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^\infty e^{(it-\lambda)x} dx \\ &= \lambda \left[\frac{e^{(it-\lambda)x}}{it-\lambda} \right]_0^\infty \\ &= \frac{\lambda}{\lambda - it}, \text{ with } \lambda = 0.5, \quad \phi_{X_i}(t) = \frac{1}{1 - it}. \end{aligned}$$

$$\text{(ii)} \quad \text{Since } X_i \text{'s are independent from each other, } \phi_{S_{36}}(t) = \prod_{i=1}^{36} \phi_{X_i}(t) = \left(\frac{1}{1 - it \cdot 0.5} \right)^{36}.$$

This is Gamma-distribution $T'(36, 0.5)$.

$$\text{(iii)} \quad X_i \sim \text{Exp}(0.5), \quad E(X_i) = \frac{1}{0.5} = 2, \quad \text{var}(X_i) = \frac{1}{0.5^2} = 4.$$

$$\text{Based on independence, } E(S_{36}) = \sum_{i=1}^{36} E(X_i) = 72, \quad \text{var}(S_{36}) = \sum_{i=1}^{36} \text{var}(X_i) = 144.$$

$$P(S_{36} > 80) = P\left(\frac{S_{36} - E(S_{36})}{\sqrt{\text{var}(S_{36})}} > \frac{80 - 72}{\sqrt{144}}\right) = P(Z > \frac{80 - 72}{12}) = P(Z > \frac{2}{3}) \approx 0.2525.$$

Ex 2.

$$\text{(i)} \quad \text{Define } X_i \text{ as resistance of } i\text{-th resistor, } E(X_i) = \mu = 100, \text{var}(X_i) = \sigma^2 = 4, \quad \bar{X}_{64} = \frac{1}{64} \sum_{i=1}^{64} X_i.$$

$$E(\bar{X}_{64}) = E\left(\frac{1}{64} \sum_{i=1}^{64} X_i\right) = \frac{1}{64} \sum_{i=1}^{64} E(X_i) = \frac{1}{64} \times 64 \times 100 = 100.$$

$$\text{Based on independence, } \text{var}(\bar{X}_{64}) = \text{var}\left(\frac{1}{64} \sum_{i=1}^{64} X_i\right) = \left(\frac{1}{64}\right)^2 \sum_{i=1}^{64} \text{var}(X_i) = \left(\frac{1}{64}\right)^2 \times 64 \times 4 = \frac{1}{16}.$$

(ii) With Chebyshev's inequality:

$$P(\bar{X}_{64} > 100.5 \text{ or } \bar{X}_{64} < 99.5) = P(|\bar{X}_{64} - 100| > 0.5) = P(|\bar{X}_{64} - E(\bar{X}_{64})| > 0.5) \leq \frac{\text{var}(\bar{X}_{64})}{(0.5)^2} = \frac{1}{16} \times 4 = \frac{1}{4}.$$

$$\begin{aligned} \text{(iii)} \quad P(\bar{X}_{64} > 100.5 \text{ or } \bar{X}_{64} < 99.5) &= P(\bar{X}_{64} > 100.5) + P(\bar{X}_{64} < 99.5) = P(Z > \frac{100.5 - E(\bar{X}_{64})}{\sqrt{\text{var}(\bar{X}_{64})}}) + P(Z < \frac{99.5 - E(\bar{X}_{64})}{\sqrt{\text{var}(\bar{X}_{64})}}) \\ &= P(Z > 2) + P(Z < -2) \quad (\text{with symmetry of normal distribution}) \\ &= 2 \cdot P(Z > 2) \approx 0.0456. \end{aligned}$$

Ex 3. Hint: For $X \sim \text{Unif}(0, \frac{1}{n})$, $p_X(x) = \begin{cases} n & \text{if } x \in [0, \frac{1}{n}] \\ 0 & \text{otherwise} \end{cases}$.

(i) Since $Y_n = nX_n = g(X_n)$, $g(y) = ny$, $g'(y) = \frac{y}{n}$, $\frac{dg^{-1}(y)}{dy} = \frac{d\frac{y}{n}}{dy} = \frac{1}{n}$.

$$P_{Y_n}(y) = P_{X_n}(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right| = P_{X_n}\left(\frac{y}{n}\right) \cdot \frac{1}{n} = \begin{cases} n \cdot \frac{1}{n} = 1 & \text{if } \frac{y}{n} \in [0, \frac{1}{n}], y \in [0, 1] \\ 0 & \text{otherwise.} \end{cases}$$

$$(ii) F_{Y_n}(y) = \int_{-\infty}^y f_{Y_n}(t) dt = \begin{cases} 0 & \text{if } y < 0 \\ y & \text{if } 0 \leq y \leq 1 \\ 1 & \text{if } y > 1 \end{cases}$$

(iii) $F_Y(y) = \lim_{n \rightarrow \infty} F_{Y_n}(y) = F_{Y_n}(y)$, since $F_{Y_n}(y)$ not depend on n .

(iv) From (iii): $\lim_{n \rightarrow \infty} F_{Y_n}(y) = F_Y(y)$, so $Y_n \xrightarrow{d} Y \sim \text{Unif}(0, 1)$.

Ex 4.

$$(i) L(\theta | x) = \prod_{i=1}^n P(x_i | \theta) = \left(\frac{1}{\theta}\right)^n \prod_{i=1}^n \chi_{[0, \theta]}(x_i) = \begin{cases} \left(\frac{1}{\theta}\right)^n & \text{if } \forall x_i \in [0, \theta] \\ 0 & \text{otherwise.} \end{cases}$$

(ii) From (i): $L(\theta | x) = \left(\frac{1}{\theta}\right)^n$ if $\forall x_i \in [0, \theta]$, i.e. $\theta \geq x_{\max}$.

As θ increases, $L(\theta | x)$ monotonically decreases, meaning when θ achieves minimum value, $L(\theta | x)$ achieves maximum value, $\hat{\theta}_{MLE} = \max \{X_1, X_2, \dots, X_n\} = X_{\max}$.

(iii) $E(\hat{\theta}) = E(X_{\max})$, to compute $E(X_{\max})$, consider pdf and cdf of $X_{\max}$.

$$F_{X_{\max}}(t) = P(X_{\max} \leq t) = P(X_1 \leq t, \dots, X_n \leq t) = \prod_{i=1}^n P(X_i \leq t) = \prod_{i=1}^n \int_0^t p(x_i = t) dt = \prod_{i=1}^n \frac{t}{\theta} = \left(\frac{t}{\theta}\right)^n.$$

$$f_{X_{\max}}(t) = F_{X_{\max}}'(t) = \left(\left(\frac{t}{\theta}\right)^n\right)' = \frac{nt^{n-1}}{\theta^n}.$$

$$E(\hat{\theta}) = E(X_{\max}) = \int_0^\theta t \cdot f_{X_{\max}}(t) dt = \int_0^\theta \frac{nt^{n-1}}{\theta^n} dt = \frac{n}{n+1} \theta \neq \theta. \hat{\theta}_{MLE} \text{ is not an unbiased estimator.}$$